

CEUR-WS Analysis



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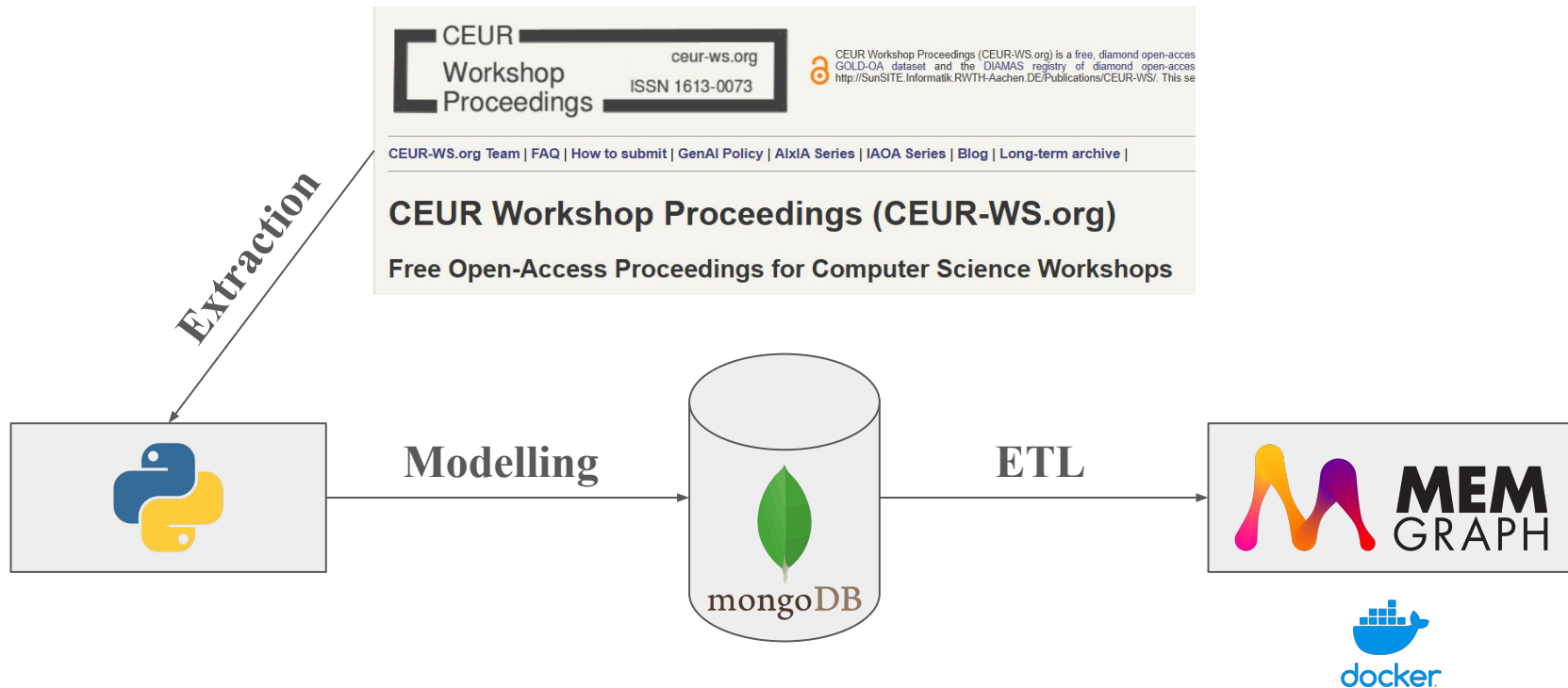
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Project Objective

Development of an automated **Data Analysis Pipeline** capable of processing academic publication data to extract, organize, and analyze information.

- Extraction of Papers' Information from their PDF located at CEUR used for the Construction of a *Scalable and Queryable Dataset*.
- Extension of the *Existing Scraper*.
- Construction of a *Large Graph Database* from the Extracted Information.

Architecture



Implementation: Parser

Smoke Screens and Scapegoats: The Reality of General Data Protection Regulation Compliance – Privacy and Ethics in the Case of Replika AI

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Abstract

Currently artificial intelligence (AI)-enabled chatbots are capturing the hearts and imaginations of the public at large. Chatbots that users can build and personalize, as well as pre-designed avatars ready for users' selection, all of these are on offer in applications to provide social companionship, friends and even love. These systems, however, have demonstrated challenges on the privacy and ethics front. This paper takes a critical approach towards examining the intricacies of these issues within AI companion services. We chose Replika as a case and employed close reading to examine the service's privacy policy. We additionally analyze articles from public media about the company and its practices to gain insight into the trustworthiness and integrity of the information provided in the policy. The aim is to ascertain whether seeming General Data Protection Regulation (GDPR) compliance equals reliability of required information, or whether the area of GDPR compliance in itself is one riddled with ethical challenges. The paper contributes to a growing body of scholarship on ethics and privacy related matters in the sphere of social chatbots. The results reveal that despite privacy notices, data collection practices might harvest personal data without users' full awareness. Cross-textual comparison reveals that privacy notice information does not fully correspond with other information sources.

Keywords

Social Chatbots, Privacy Policies, Ethics, AI Companions, Data Protection

1. Introduction

Artificial Intelligence (AI) systems are becoming ubiquitous in everyday life. Reliance on smart technologies for personal contexts, industry, healthcare and education are increasing with the development of even more AI-products. Recent themes that have emerged in the development of AI technologies include affective computing, emotional AI, and empathetic AI [1]. At the heart of these initiatives is the goal of inciting affective emotional relationships between humans and technology [2]. AI capable of conversation and interaction with users through voice, text, image, and video have managed to penetrate the social sphere. This is particularly afforded by design features that embed emotional characteristics into the fabrics of the technology. Such systems are designed and trained to recognize, simulate and elicit human emotions and can serve the function of social interaction [3]. These types of AI that interact with users on an emotional and personal level are commonly referred to as social chatbots.

Social chatbots mimic human conversations via Natural Language Processing (NLP) and simulate human characteristics such as speech, text, gestures, and facial expressions [4]. They are commonly manifested as text-based chatbots, digital avatars, and social robots designed for conversation and interaction. Recently, the development of more humanlike chatbots has become possible with

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Implementation: Parser

Problem: Citations are often in Different Conventions.

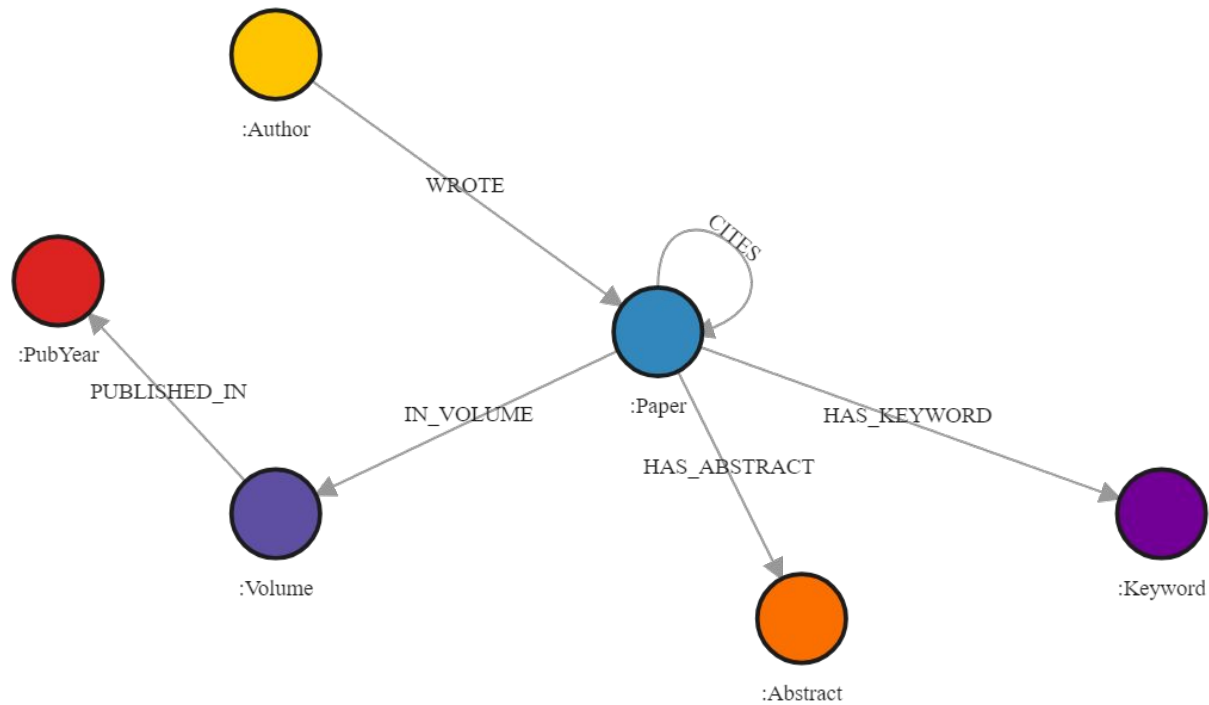
Solution: Detect a Sequence of English Words which Mark where the Title is.

- [1] M.-H. Huang, R. Rust, V. Maksimovic, The Feeling Economy: Managing in the Next Generation of Artificial Intelligence (AI), *California Management Review* 61(4) (2019) 43–65. <https://doi.org/10.1177/0008125619863436>.
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- 2. Koide, Seiji, and Hideaki Takeda: RDFization of Japanese Electronic Dictionaries and LOD. *2nd Workshop on Linked Data in Linguistics (LDL-2013)*, pp.64 –69, Association for Computational Linguistics (2013)

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Graph Structure



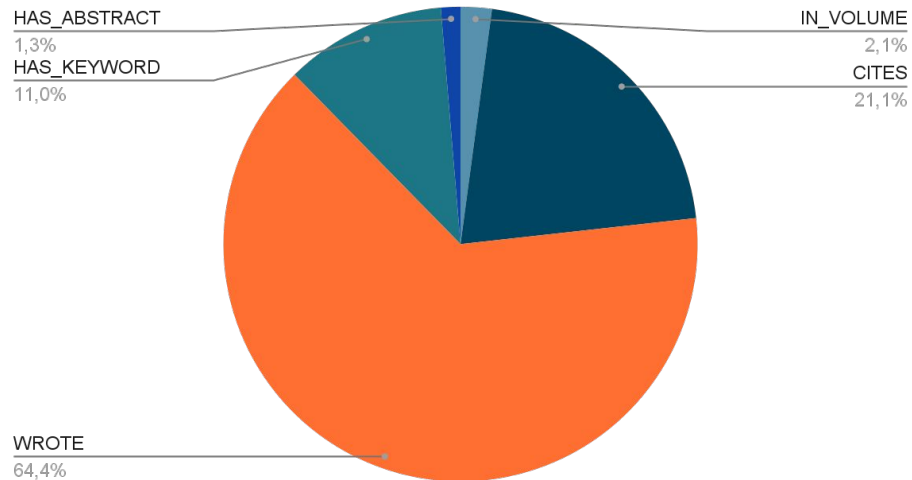
Results • General Stats

Nodes: 758.843

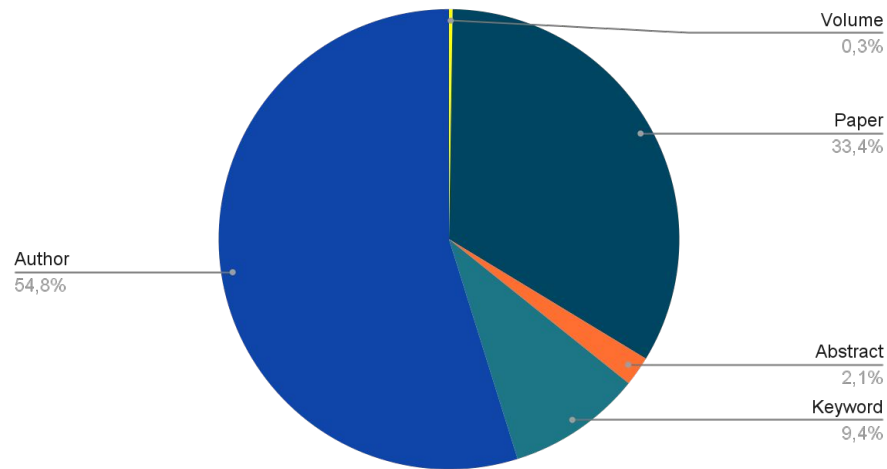
Edges: 1.190.000

Storage: 857 MB

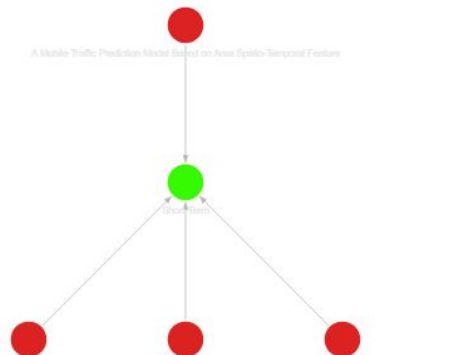
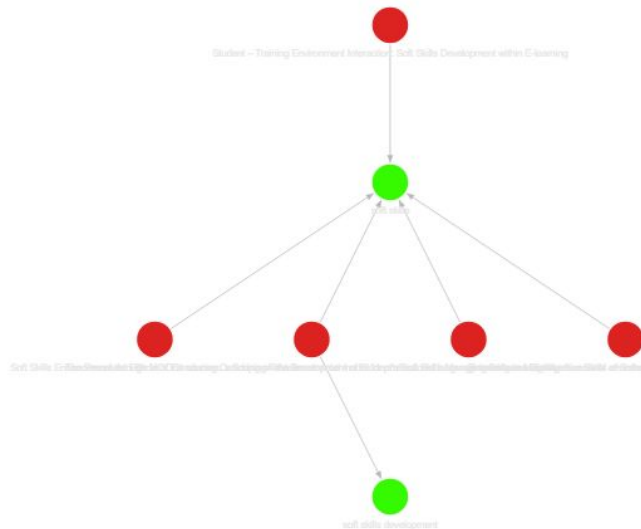
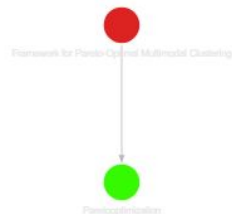
Relationships



Nodes



Results • Group by Keywords

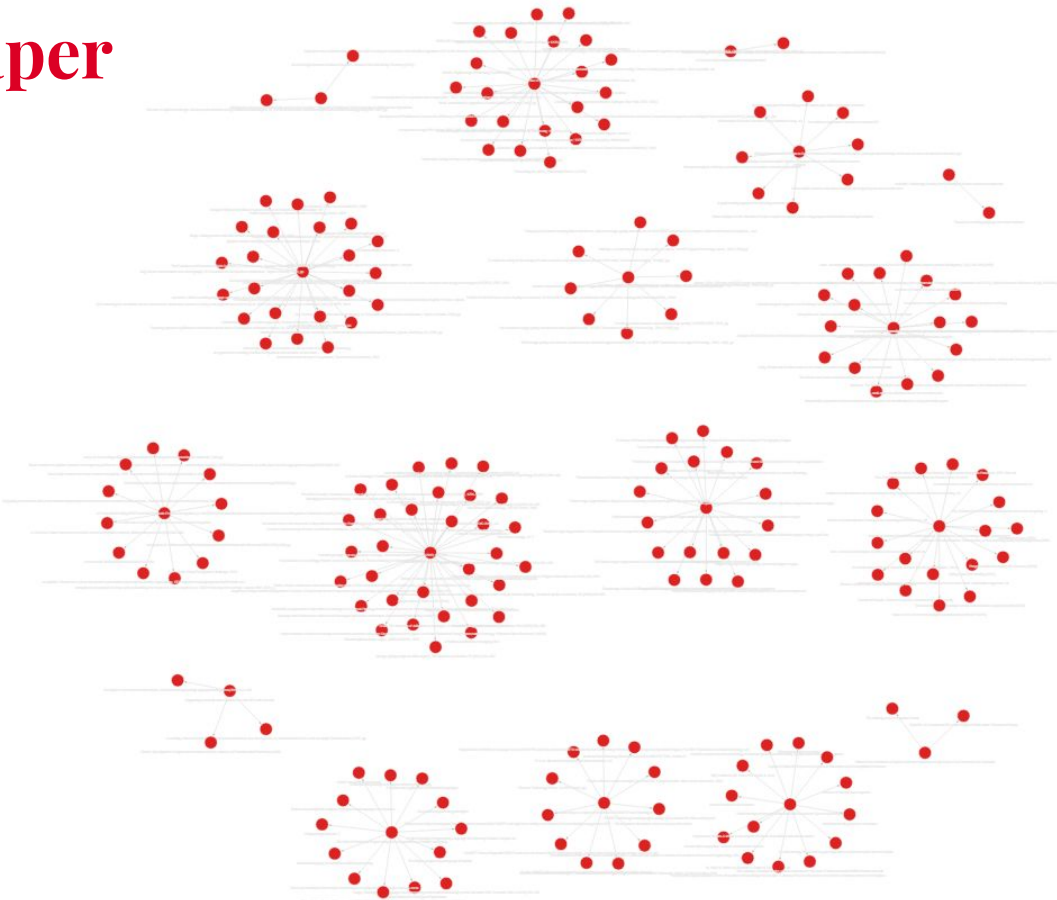


```
MATCH p=( )-[ :HAS_KEYWORD ]->(k:Keyword)
RETURN p LIMIT 500
```

● Keywords
● Papers

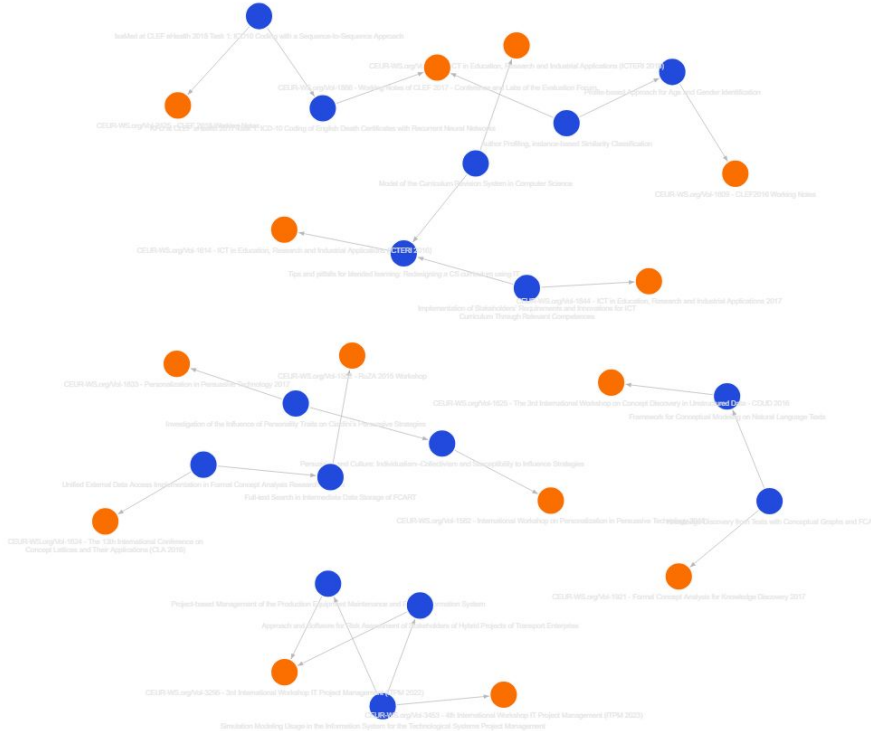
Results • Group by Paper

```
MATCH p=()-[:CITES]-(target:Paper)  
RETURN p LIMIT 200
```



Results • Connected Volumes

● Volumes
● Papers



```
MATCH (v1:Volume)<-[i:IN_VOLUME]-(p1:Paper)-[c:CITES]
      |> (p2:Paper)-[i2:IN_VOLUME]->(v2:Volume)
WHERE v1 <> v2
RETURN v1 AS source, v2 AS target
      , count(*) AS num_citazioni, i
      , p1, c, p2, i2
LIMIT 10;
```

isolated_volumes	connected_volumes	total_volumes
1,624	286	1,910

Results • Most Used Keywords

```
MATCH (p:Paper)-[:HAS_KEYWORD]->(k:Keyword),
      (p)-[:IN_VOLUME]->(v:Volume)-[:PUBLISHED_IN]->(y:PubYear)
WITH y.year AS year, k.name AS keyword, count(*) AS freq
ORDER BY year, freq DESC
WITH year, collect({k: keyword, freq: freq})[0..3] AS top_keywords
UNWIND top_keywords AS kw
RETURN year, kw.k AS keyword, kw.freq AS frequency
ORDER BY year, frequency DESC;
```

29	2021	Ontology	63	▼
30	2021	BERT	63	▼
31	2022	MachineLearning	69	▼
32	2022	BERT	55	▼
33	2022	NaturalLanguageProcessing	51	▼
34	2023	MachineLearning	62	▼
35	2023	NaturalLanguageProcessing	46	▼
36	2023	LargeLanguageModels	43	▼
37	2024	LargeLanguageModels	177	▼
38	2024	NaturalLanguageProcessing	101	▼
39	2024	MachineLearning	73	▼
40	2025	LargeLanguageModels	121	▼
41	2025	machine learning	49	▼
42	2025	LLM	44	▼

Limitations and Future Work

Data Consistency and Upgrading:

- Improvement of Correct Data Parsing
- Extend the Parsing and Nodes to Chapters, Number of Chapters, and Other Info

Web Interface w/ LLM:

- An Easy way to Use the Tool, Improving Search and Automated Answer

Data Analysis Extension Beyond CEUR:

- Search for Related Papers in other Websites for Academic Papers
 - OpenAlex <https://docs.openalex.org/>



**Thank You
For Your Attention**